Fundamentals of Linear Control:
A Concise Approach

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Chapter 6: Controller Design

linearcontrol.info/fundamentals

6.1 Second-Order Systems

Characteristic equation

$$s^2 + 2\,\zeta\,\omega_n s + \omega_n^2 = 0, \qquad \omega_n > 0,$$

- ζ: damping ratio
- ω_n: natural frequency

Roots

$$s^2 + 2\,\zeta\,\omega_n s + \omega_n^2 = (s - s_1)(s - s_2), \qquad s_1 = -\omega_n(\zeta + \sqrt{\zeta^2 - 1}), \qquad s_2 = -\omega_n(\zeta - \sqrt{\zeta^2 - 1})$$

Real roots

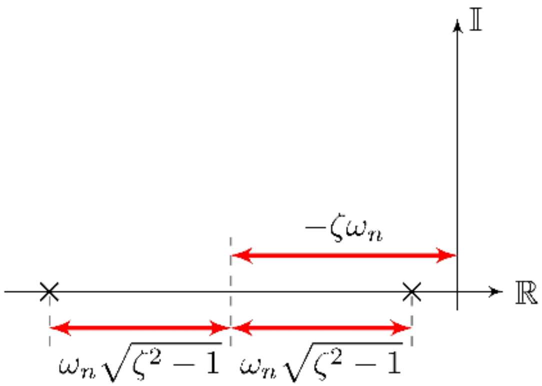
$|\zeta| \geq 1$

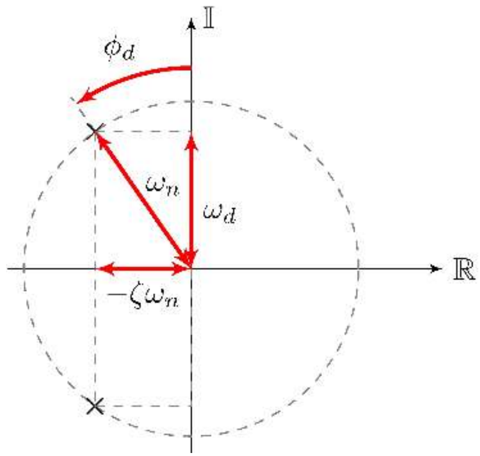
Complex roots

$|\zeta| < 1$

Second-Order Systems

real roots, $|\zeta| \geq 1$ complex roots, $|\zeta| < 1$





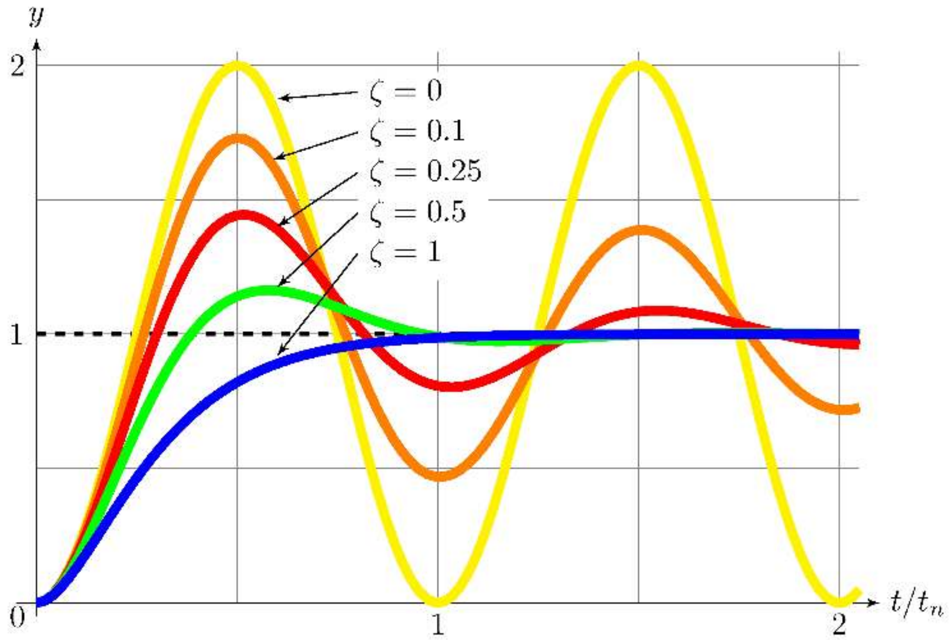
Roots

$$s = -\omega_n(\zeta \pm \sqrt{\zeta^2 - 1}), \qquad \omega_d = \omega_n\sqrt{1-\zeta^2}, \qquad \phi_d = \sin^{-1}\frac{\zeta\omega_n}{\omega_n} = \tan^{-1}\frac{\zeta\,\omega_n}{\omega_d} = \sin^{-1}\zeta$$

Asymptotic stability

$\zeta > 0$

Second-Order Systems

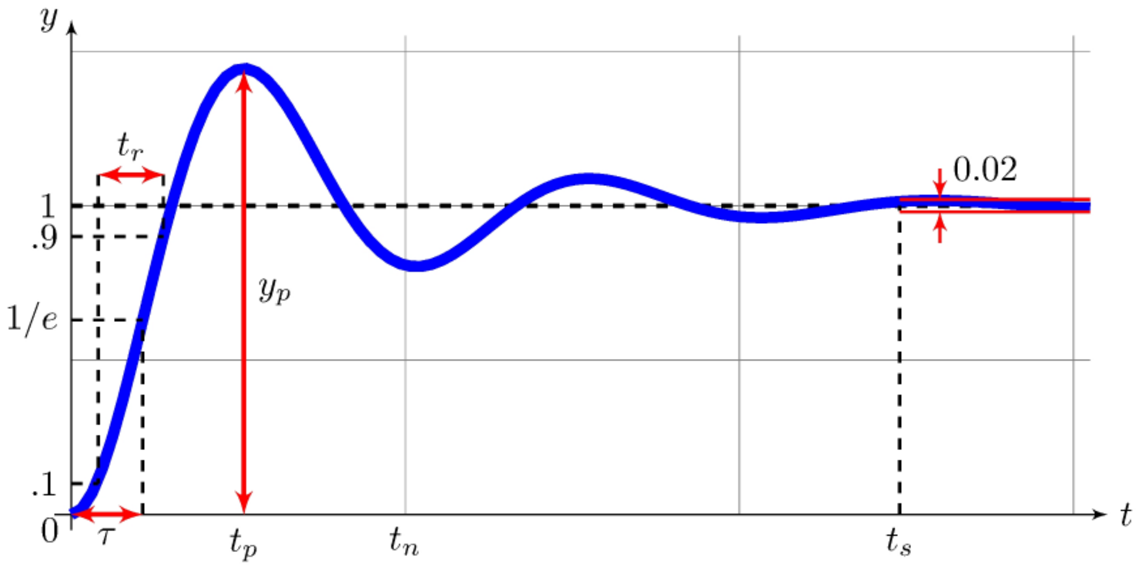


Step response

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \qquad y(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \pi/2 - \phi_d), \quad t \geq 0$$

Second-Order Systems

Step response

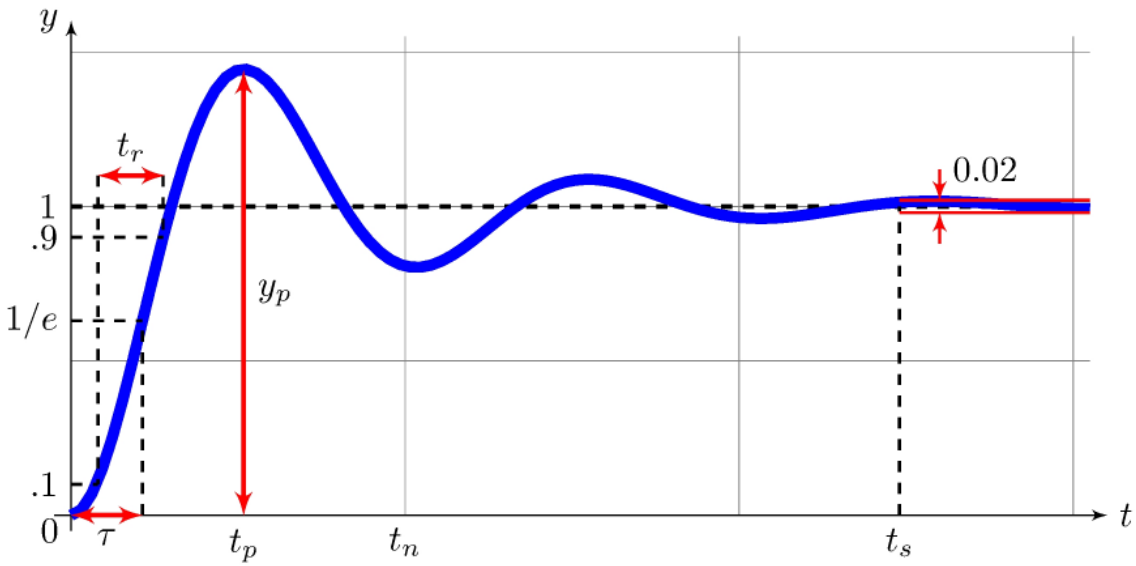


Time-constant and Rise-time

$$\frac{t_r}{t_n} \approx 0.16 + 0.14\,\zeta + 0.24\,\zeta^3, \qquad \frac{\tau}{t_n} \approx 0.19 + 0.1\,\zeta + 0.054\,\zeta^3, \qquad t_n = \frac{2\pi}{\omega_n}$$

Second-Order Systems

Step response



Overshoot and Settling-time

$$t_p = \frac{\pi}{\omega_d}, \qquad y_p = 1 + e^{-\zeta\pi/\sqrt{1-\zeta^2}}, \qquad t_s = \frac{\log(50)}{\zeta\,\omega_n} \approx \frac{3.9}{\zeta\,\omega_n}$$

Example: car model with integral feedback

Open-loop

$$G(s) = \frac{\frac{p}{m}}{s + \frac{b}{m}}, \qquad K(s) = \frac{K_i}{s}$$

Closed-loop

$$H(s) = \frac{K(s)G(s)}{1 + K(s)G(s)} = \frac{\frac{p}{m} K_i}{s^2 + \frac{b}{m} s + \frac{p}{m} K_i}$$

Second-order

$$\omega_n = \sqrt{\frac{p}{m} K_i}, \qquad \zeta = \frac{b/m}{2\sqrt{(p/m)\, K_i}} = \frac{b}{2\sqrt{p\, m\, K_i}}$$

Example: car model with proportional-integral feedback

Open-loop

$$G(s) = \frac{\frac{p}{m}}{s + \frac{b}{m}}, \qquad K(s) = K_p + \frac{K_i}{s}$$

Closed-loop

$$H(s) = \frac{K(s)G(s)}{1 + K(s)G(s)} = \frac{\frac{p}{m}(K_p s + K_i)}{s^2 + (\frac{b}{m} + \frac{p}{m}K_p)s + \frac{p}{m}K_i}$$

Second-order

$$\omega_n = \sqrt{\frac{p}{m}K_i}, \qquad \zeta = \frac{b/m + p/mK_p}{2\sqrt{(p/m)\,K_i}} = \frac{b + pK_p}{2\sqrt{p\,m\,K_i}}$$

Example: simple pendulum

Open-loop (stable equilibrium)

$$G_0(s) = \frac{1/J_r}{s^2 + (b/J_r)s + m\,g\,r/J_r}$$

Second-order

$$\omega_{n_{\mathrm{ol}}} = \sqrt{\frac{mgr}{J_r}}, \qquad \zeta_{\mathrm{ol}} = \frac{b}{2\sqrt{J_r mgr}}$$

Open-loop (unstable equilibrium)

$$G_\pi(s) = \frac{1/J_r}{s^2 + (b/J_r)s - m\,g\,r/J_r}$$

Example: simple pendulum

Open-loop

$$G_\pi(s) = \frac{1}{J_r s^2 + bs - m\,g\,r}$$

Proportional feedback (unstable equilibrium)

$$H_\pi(s) = \frac{KG_\pi(s)}{1 + KG_\pi(s)} = \frac{K}{J_r s^2 + bs + K - m\,g\,r},$$

$$\omega_{n_\pi} = \sqrt{\frac{K - m\,g\,r}{J_r}}, \qquad \zeta_\pi = \frac{b}{2\sqrt{J_r(K - m\,g\,r)}}$$

Closed-loop stability

$K > mgr$, for example

$$K = 3mgr \qquad \Longrightarrow \qquad \omega_{n_\pi} = \sqrt{2}\,\omega_{n_{\mathrm{ol}}}, \qquad \zeta_\pi = \frac{1}{\sqrt{2}}\zeta_{\mathrm{ol}}$$

Example: simple pendulum

Open-loop

$$G_0(s) = \frac{1}{J_r s^2 + bs + m\,g\,r}, \qquad \omega_{n_{\text{ol}}} = \sqrt{\frac{mgr}{J_r}}, \qquad \zeta_{\text{ol}} = \frac{b}{2\sqrt{J_r mgr}}$$

Proportional feedback (stable equilibrium)

$$H_0(s) = \frac{KG_0(s)}{1 + KG_0(s)} = \frac{K}{J_r s^2 + bs + K + m\,g\,r},$$

$$\omega_{n_0} = \sqrt{\frac{K + m\,g\,r}{J_r}}, \qquad \zeta_0 = \frac{b}{2\sqrt{J_r(K + m\,g\,r)}}$$

Closed-loop stability

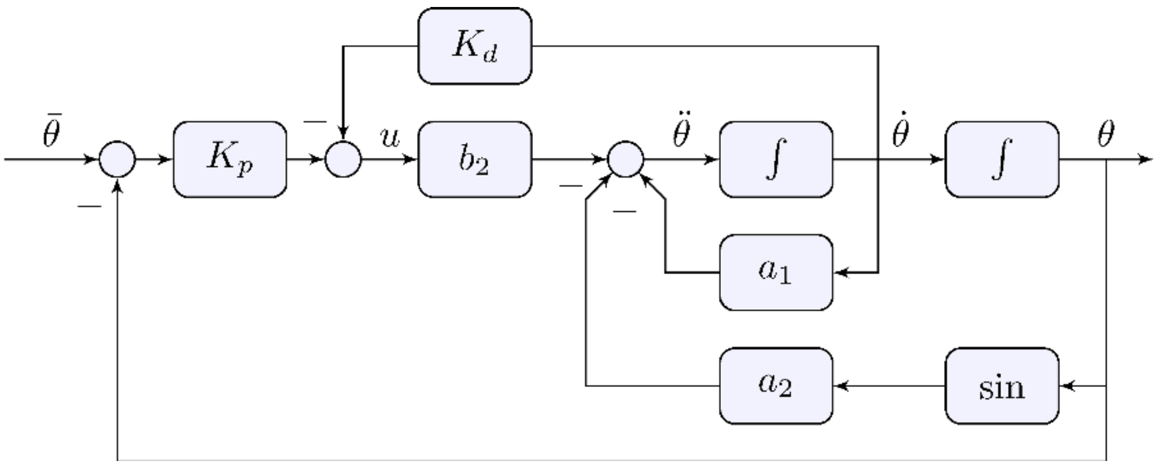
$K > -mgr$, for example

$$K = 3mgr \qquad \implies \qquad \omega_{n_0} = 2\,\omega_{n_{\text{ol}}}, \qquad \zeta_0 = \frac{1}{2}\zeta_{\text{ol}}$$

6.2 Derivative Action

Example: simple pendulum

$$u(t) = K_p\left(\bar{\theta}(t) - \theta(t)\right) - K_d\,\dot{\theta}(t)$$

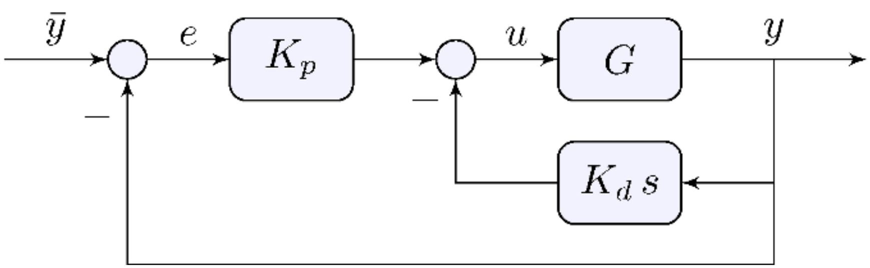


Proportional-derivative feedback

$$u(t) = K_p\,(\bar{y}(t) - y(t)) - K_d\,\dot{y}(t)$$

Frequency-domain

$$U(s) = K_p E(s) - K_d\,sY(s), \qquad E(s) = \bar{Y}(s) - Y(s)$$

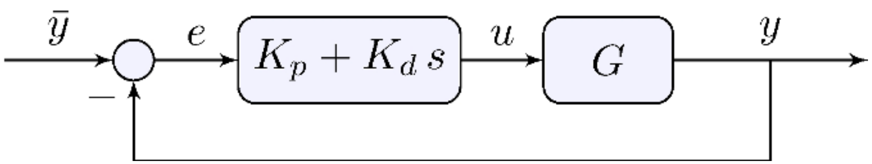


Proportional-derivative feedback

$$u(t) = K_p\, e(t) + K_d\, \dot{e}(t), \qquad e(t) = \bar{y}(t) - y(t)$$

Frequency-domain

$$U(s) = (K_p + sK_d)E(s), \qquad E(s) = \bar{Y}(s) - Y(s)$$



- Same if $\bar{y}$ is constant!
- If not, extra zero

Example: simple pendulum

Unstable equilibrium

$$H_\pi(s) = \frac{(K_p + K_d s) G_\pi(s)}{1 + (K_p + s K_d) G_\pi(s)} = \frac{K_p + K_d s}{J_r s^2 + (K_d + b)s + (K_p - m\,g\,r)},$$

$$\omega_{n_\pi} = \sqrt{\frac{K_p - m\,g\,r}{J_r}}, \qquad \zeta_\pi = \frac{K_d + b}{2\sqrt{J_r(K_p - m\,g\,r)}}$$

- Choose K_p to set desired natural frequency, e.g.

$$K = 3mgr \qquad \implies \qquad \omega_{n_\pi} = \sqrt{2}\,\omega_{n_{\mathrm{ol}}}$$

- Choose K_d to set desired damping ratio, e.g. $\zeta_\pi = \frac{\sqrt{2}}{2} \approx 0.7$

$$K_d = 2\,\zeta_\pi \sqrt{J_r(K_p - m\,g\,r)} - b = 2\sqrt{J_r m\,g\,r} - b$$

- Similarly around stable equilibrium

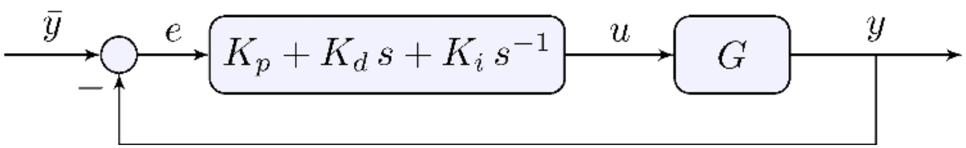
6.3 Proportional-Integral-Derivative Control

Proportional-integral-derivative feedback

$$u(t) = K_p\, e(t) + K_d\, \dot{e}(t) + K_i \int_0^t e(\tau)\, d\tau, \qquad e(t) = \bar{y}(t) - y(t)$$

Frequency-domain

$$U(s) = \left(K_p + K_d\, s + K_i s^{-1}\right) E(s) = \frac{K_d\, s^2 + K_p\, s + K_i}{s} E(s), \qquad E(s) = \bar{Y}(s) - Y(s)$$



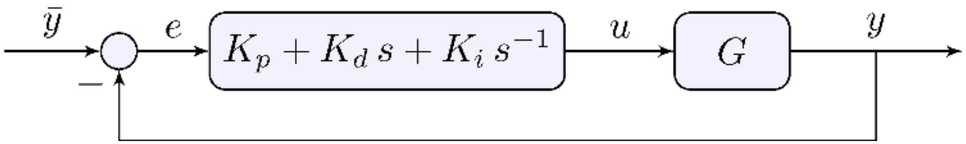
- integral action: asymptotic tracking of constant reference
- derivative action: better "damping"
- Extensive literature on properties of PID control

Proportional-integral-derivative feedback

$$u(t) = K_p\, e(t) + K_d\, \dot{e}(t) + K_i \int_0^t e(\tau)\, d\tau, \qquad e(t) = \bar{y}(t) - y(t)$$

Frequency-domain

$$U(s) = \left(K_p + K_d\, s + K_i s^{-1}\right) E(s) = \frac{K_d\, s^2 + K_p\, s + K_i}{s} E(s), \qquad E(s) = \bar{Y}(s) - Y(s)$$

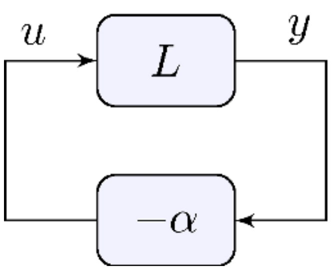


- Transfer-function is not proper: not realizable!
- If G is second-order then closed-loop transfer-functions are third-order
- Calculation of the characteristic roots is *already* too complicated

6.4 Root-Locus

The basic idea

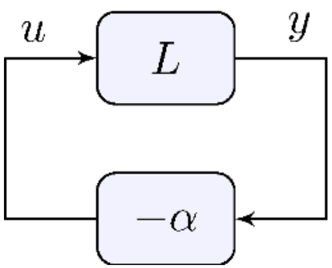
- Can we quickly *sketch* the location of the closed-loop poles without calculating them?
- Root *locus* $=$ location of the roots
- Set of simple rules for sketching the root-locus of the feedback system



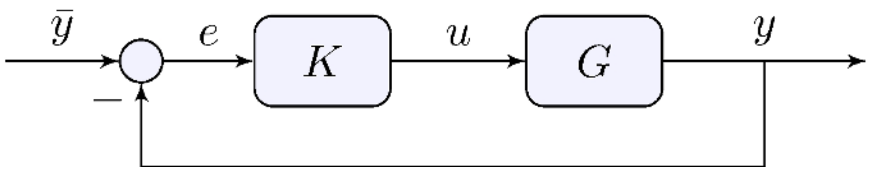
as a function of $\alpha > 0$

- Make your feedback system look like it and apply the rules
- L is the *loop transfer-function*

Feedback diagram for root-locus



Example



Closed-loop poles

$$1 + \alpha L(s) = 0, \qquad K(s) = \alpha C(s), \qquad L(s) = C(s)G(s)$$

Root-locus rules

Key observation

$$L(s) = -\frac{1}{\alpha} < 0 \qquad \Longrightarrow \qquad \angle L(s) = \pi$$

Rational loop transfer-function

$$L = \beta \frac{N_L}{D_L} = \frac{\beta\,(s - z_1)(s - z_2)\cdots(s - z_m)}{(s - p_1)(s - p_2)\cdots(s - p_n)}$$

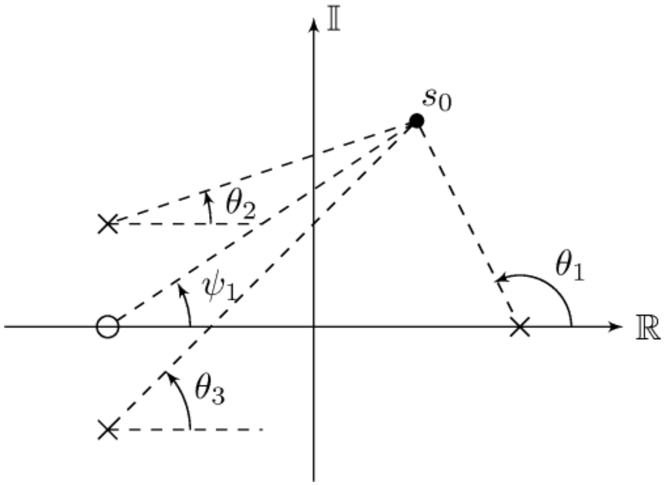
then

$$\angle L = \angle \beta + \angle N_L - \angle D_L = \angle \beta + \sum_{i=1}^{m} \angle (s - z_i) - \sum_{k=1}^{n} \angle (s - p_k)$$

Root-locus rules

Phase can be calculated graphically

$$\angle L(s_0) = \angle\beta + \sum_{i=1}^{m} \angle(s_0 - z_i) - \sum_{k=1}^{n} \angle(s_0 - p_k)$$
$$= \angle\beta + \psi_1 - \theta_1 - \theta_2 - \theta_3$$



Root-locus rules

Assumption

L rational and strictly proper with real coefficients

Rules

- Root-locus is a set of n continuous curves
- The curves:
  - begin at the poles of L

$$\lim_{\alpha\to 0} D_L(s)\,(1+\alpha L(s)) = \lim_{\alpha\to 0} D_L(s) + \alpha\beta N_L(s) = D_L(s)$$

  - end at the zeros of L

$$\lim_{\alpha\to\infty} \alpha^{-1} D_L(s)\,(1+\alpha L(s)) = \lim_{\alpha\to\infty} \alpha^{-1} D_L(s) + \beta N_L(s) = \beta N_L(s)$$

- $n - m \geq 1$ curves go to infinity

Root-locus rules

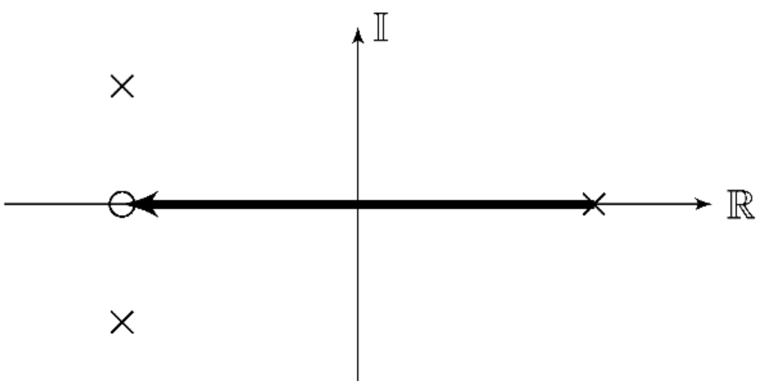
Assumption

L rational and strictly proper with real coefficients and $\beta > 0$

Rule

- $s \in \mathbb{R}$ is in the root-locus iff L has an odd number of poles and zeros with real part greater than or equal to s

Example (complex locus not shown)



- What would the $\beta < 0$ version be?

Root-locus rules

Assumption

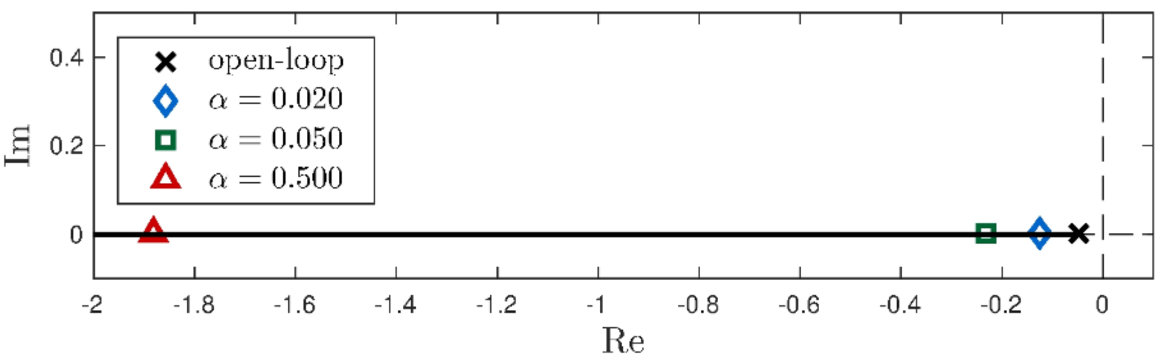
L rational and strictly proper with real coefficients and $\beta > 0$

Rule

- $s \in \mathbb{R}$ is in the root-locus iff L has an odd number of poles and zeros with real part greater than or equal to s

Example: car model under proportional feedback

$$\alpha = K_p, \qquad L(s) = \frac{\frac{p}{m}}{s + \frac{b}{m}} = \frac{3.7}{s + 0.05}$$



Root-locus rules

Assumption

L rational and strictly proper with real coefficients

Rule

- If $n > m + 1$, then $n - m$ roots converge to $n - m$ straight-line asymptotes intersecting at:

$$c = \frac{\sum_{k=1}^{n} p_k - \sum_{i=1}^{m} z_i}{n - m} \in \mathbb{R}$$

- If $\beta > 0$, the asymptotes intersect the real axis at angles

$$\phi_k = \frac{\pi + k\,2\pi}{n - m}, \quad k = 0, \cdots, n - m - 1$$

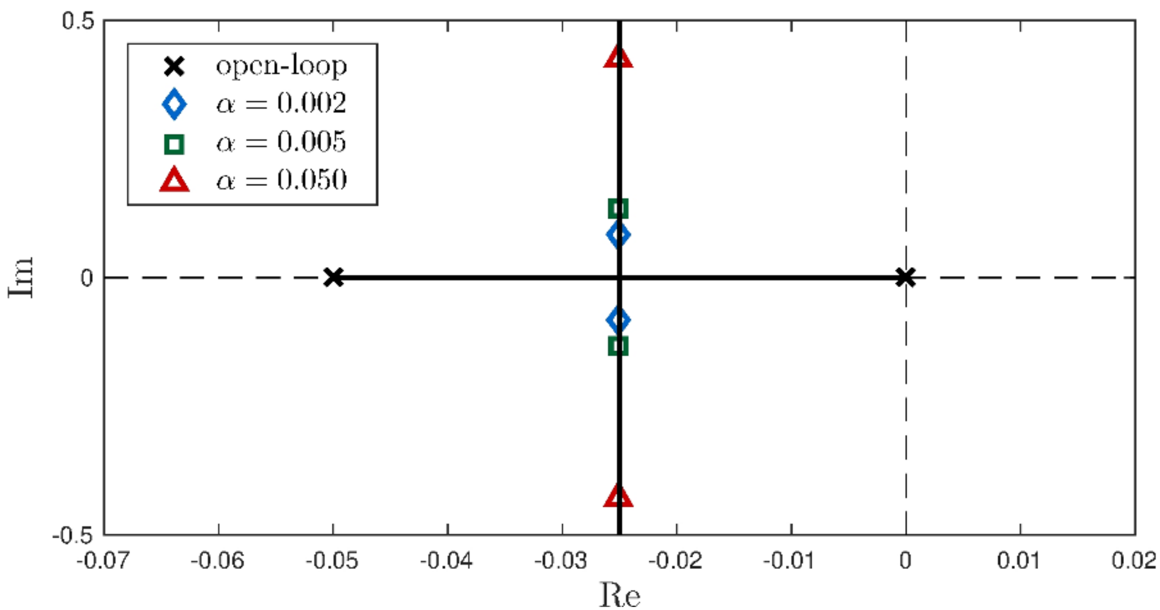
Root-locus rules

Example: car model under integral feedback

$$\alpha = K_i, \qquad L(s) = G(s)\,C(s) = \frac{3.7}{s(s+0.05)}$$

Asymptotes

$$c = \frac{-0.05 + 0}{2} = -0.025, \qquad \phi_1 = \frac{\pi}{2}, \qquad \phi_2 = \frac{3\pi}{2}$$



Root-locus rules

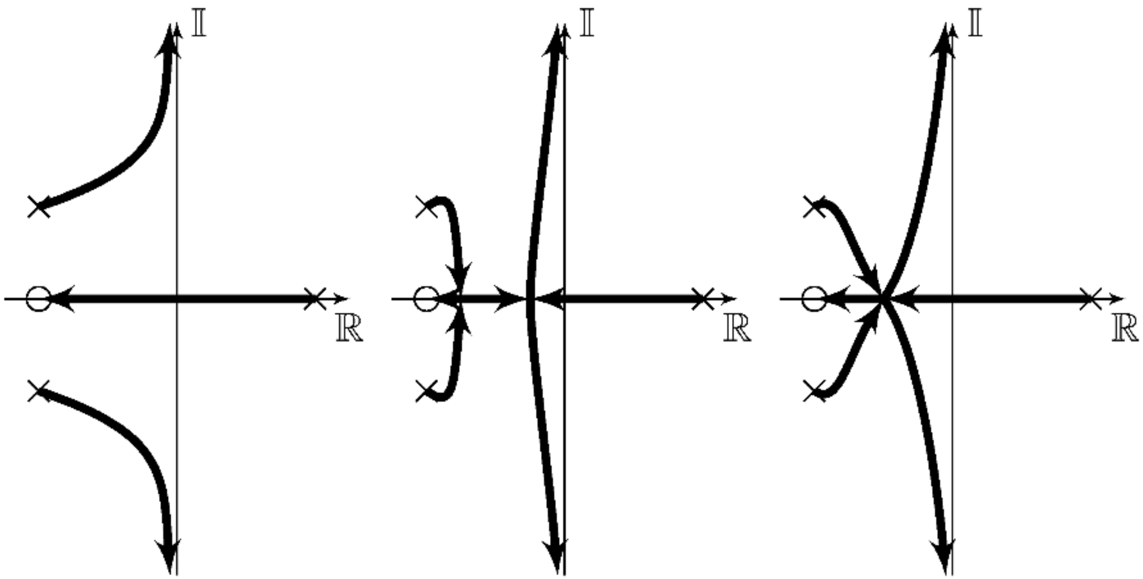
Example: model with three poles and one zero

$$p_1 = a, \qquad p_2 = -a + jb, \qquad p_2 = -a - jb, \qquad z_1 = -a$$

2 asymptotes

$$c = \frac{(-a - jb) + (-a + jb) + (a) - (-a)}{3 - 1} = 0, \qquad \phi_1 = \frac{\pi}{2}, \qquad \phi_2 = \frac{3\pi}{2}$$

Possible root-loci



Root-locus rules

Example: car model under proportional-integral feedback

$$G = \frac{\frac{p}{m}}{s + \frac{b}{m}} = \frac{3.7}{s + 0.05}, \qquad K(s) = K_p + \frac{K_i}{s} = K_p \frac{s + K_i/K_p}{s}$$

Loop transfer-function

$$\alpha = K_p, \qquad L = G\,C = \frac{3.7\,(s + K_i/K_p)}{s\,(s + 0.05)}$$

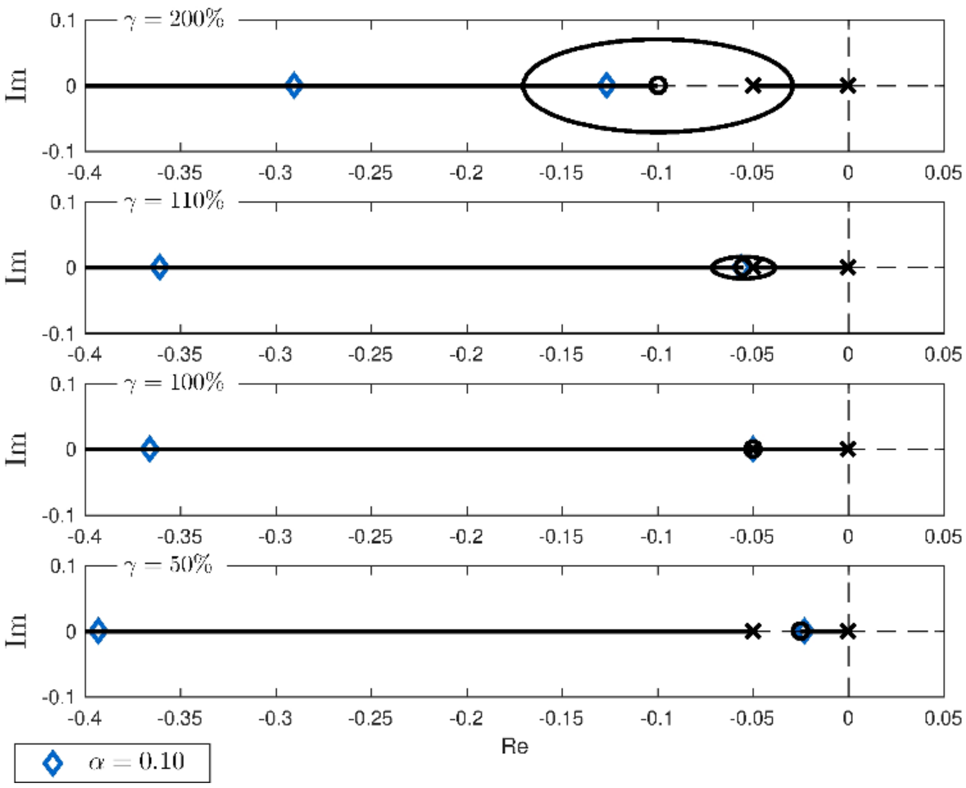
Choice of controller zero

$$\frac{K_i}{K_p} = \gamma \frac{b}{m} = \gamma\,0.05$$

- $\gamma = 1$ is a pole-zero cancellation

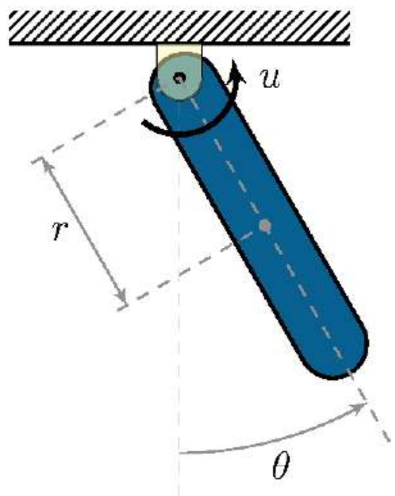
Root-locus rules

Example: car model under proportional-integral feedback



6.5 Control of the Simple Pendulum - Part I

Proportional-derivative feedback



Controller

$$C_{\mathrm{PD}}(s) = K_d\,(s+z), \qquad z = \frac{K_p}{K_d}$$

Data

$$m = 0.5\text{kg}, \qquad \ell = 0.3\text{m}, \qquad r = \ell/2 = 0.15\text{m}$$

$$b = 0\text{kg/s}, \qquad g = 9.8\text{m/s}^2, \qquad J = \frac{m\ell^2}{12} = 3.75 \times 10^{-3}\text{kg m}^2, \qquad J_r = J + mr^2 = 0.015$$

Gains

$$K_d = 3\,m\,g\,r \approx 0.21, \qquad K_p = 2\sqrt{J_r m\,g\,r} - b \approx 2.21, \qquad z \approx 10.5, \qquad C_{\mathrm{PD}}(s) \approx 0.21(s+10.5)$$

Proportional-derivative feedback

System model

$$G_\pi(s) = \frac{66.7}{(s+7)(s-7)}$$

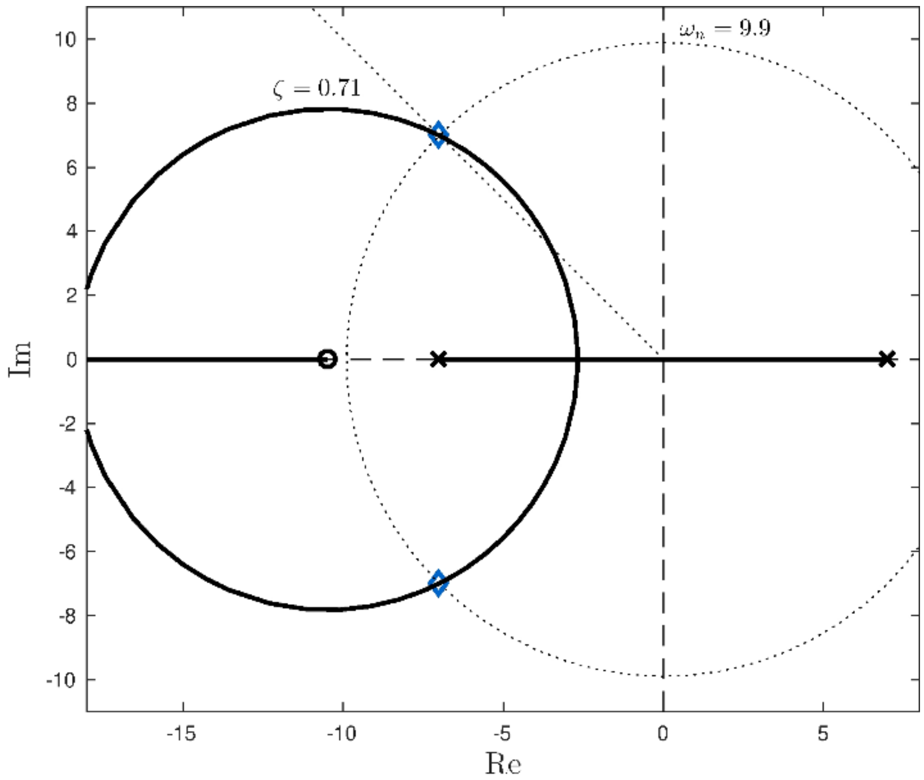
Controller

$$K(s) = C_{\text{PD}}(s) = K_d(s+z), \qquad z = \frac{K_p}{K_d} > 0$$

Root-locus

$$\alpha = K_d, \qquad L = \frac{1}{K_d} G_\pi C_{\text{PD}} = \frac{66.7(s+z)}{(s+7)(s-7)}$$

$z = 10.5$



- Diamonds: $K_d \approx 0.21 \implies \omega_{n_\pi} = \sqrt{2}\,\omega_{n_{\text{ol}}} \approx 9.90, \quad \zeta_\pi = 1/\sqrt{2} \approx 0.71$
- PD controller is not proper, hence not realizable: we need a pole
- How about a pole at the origin for some desirable integral action?

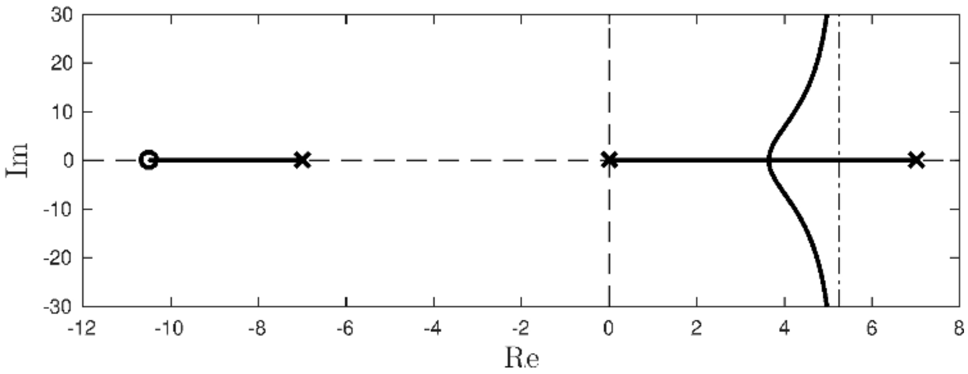
Proportional-integral feedback

Controller

$$K = K_p \frac{(s+z)}{s}$$

Root-locus

$$\alpha = K_p, \qquad L = \frac{66.7(s+z)}{s(s+7)(s-7)}$$



- No $K_p > 0$ can make closed-loop stable!

Proportional-derivative feedback plus pole

Controller

$$K(s) = K \frac{s+z}{s+p}$$

Root-locus

$$\alpha = K \qquad L = \frac{66.7(s+z)}{(s+7)(s-7)(s+p)}$$

2 asymptotes

$$c = \frac{7 - 7 + (-p) - (-z)}{n-m} = \frac{z-p}{2}, \qquad \phi_1 = \frac{\pi}{2}, \qquad \phi_2 = \frac{3\pi}{2}$$

Stability

Possible with large K only if

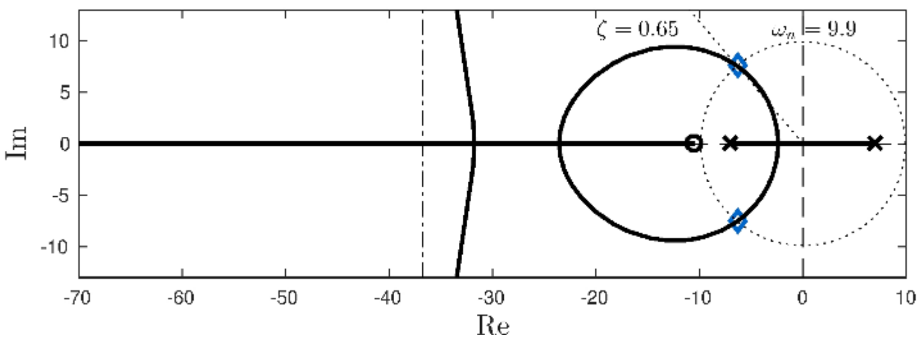
$$c < 0 \qquad \Longrightarrow \qquad p > z$$

Proportional-derivative feedback plus pole

Root-locus

$$\alpha = K \qquad L = \frac{66.7(s+z)}{(s+7)(s-7)(s+p)}$$

Choice of pole



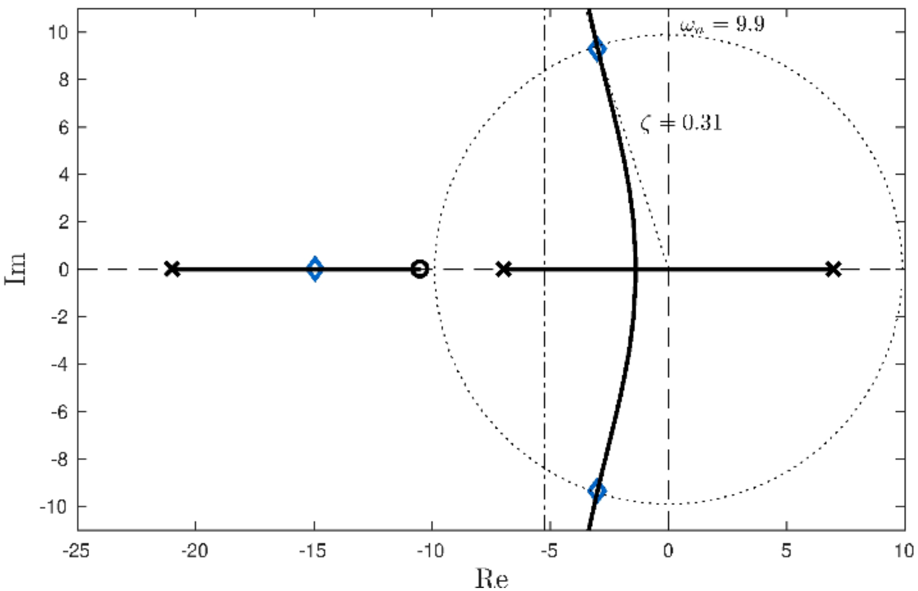
- $p = 8z \gg z \approx 10.5$
- Root-locus near origin is mostly unaffected

Proportional-derivative feedback plus pole

Root-locus

$$\alpha = K \qquad L = \frac{66.7(s+z)}{(s+7)(s-7)(s+p)}$$

Choice of pole



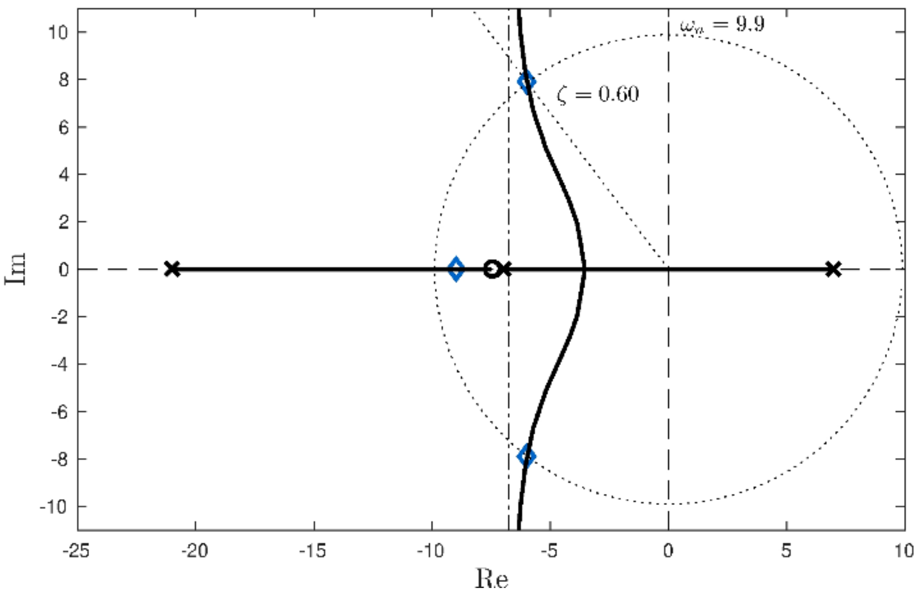
- $p = 2z > z \approx 10.5$
- Asymptote interferes with root-locus near the origin
- Low damping

Proportional-derivative feedback plus pole

Root-locus

$$\alpha = K \qquad L = \frac{66.7(s+z)}{(s+7)(s-7)(s+p)}$$

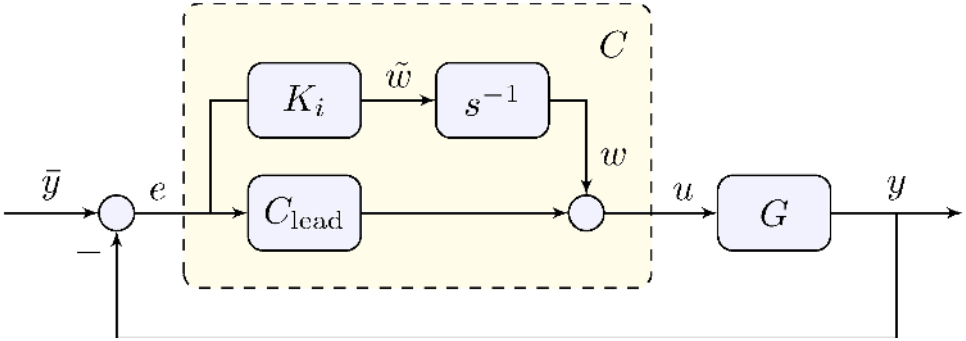
Choice of pole and adjustment of zero



- $p = 2z > \tilde{z} \approx 7.5$
- Adjusting zero improves damping
- Final controller

$$C_{\text{lead}}(s) \approx 3.83 \frac{s+7.5}{s+21}$$

Can we add integral action?



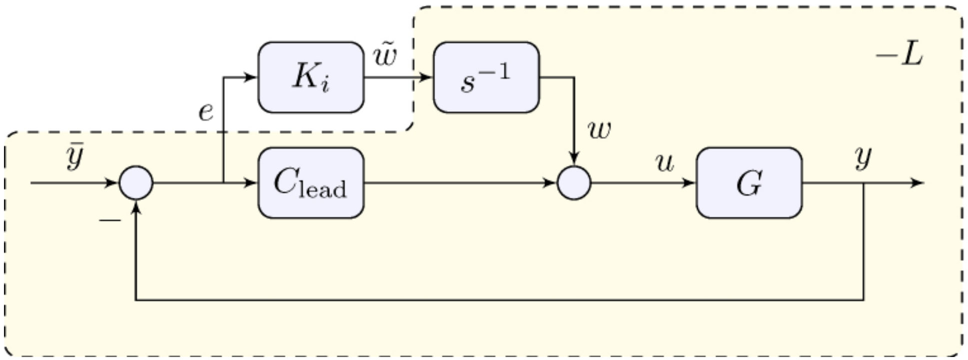
- Previous attempt failed to stabilize
- Additive approach means that stability is guaranteed for K_i small
- Structure of controller

$$C(s) = \frac{K_i}{s} + C_{\text{lead}}(s) = K \frac{s^2 + (z + K_i/K)s + pK_i/K}{s\,(s+p)}$$

- What is L?

Can we add integral action?

Root-locus analysis



Loop transfer-function

$$\alpha = K_i, \qquad L = -\frac{E}{\tilde{W}} = -s^{-1}\frac{E}{W} = s^{-1}D = \frac{s^{-1}G_\pi}{1 + G_\pi C_{\text{lead}}}$$

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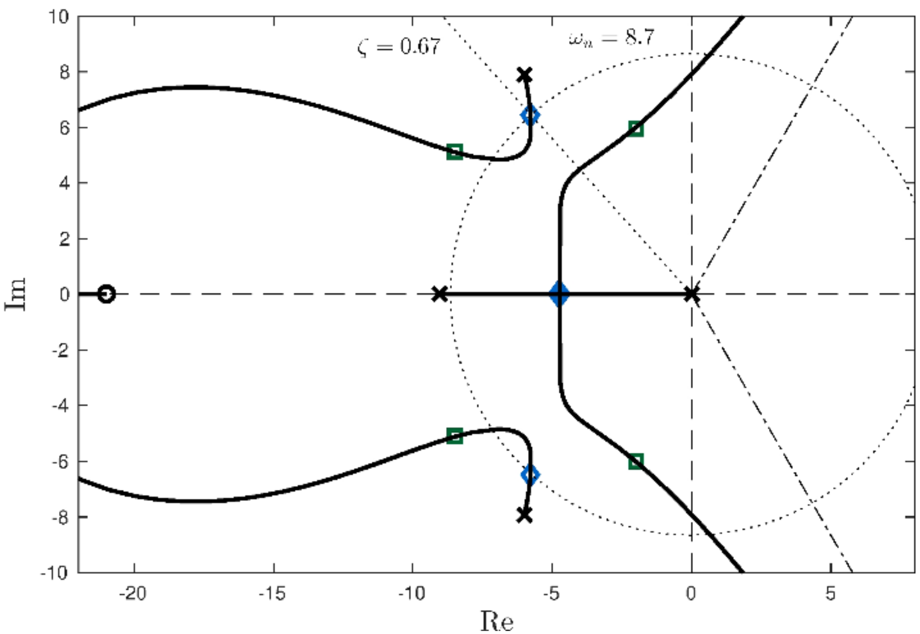
Choice of integral gain



- Diamonds: double real poles, $\omega_n \approx 8.7$
- Squares: $\omega_n \approx 9.9$, underdamped poles
- Final controller:

$$C_6(s) \approx 3.83\frac{(s + 6.86)(s + 0.96)}{s\,(s + 21)}$$

- See book for:
  - controller design around the stable equilibrium
  - direct root-locus design with integrator

Learn more

- Book website